

RAG Document Search Troubleshooting and Operations Guide

1. Render Deployment and Memory Optimization

Render free tier provides 512 MB of RAM and 1 vCPU. To prevent Out-Of-Memory (OOM) crashes, the native ONNX runtime is constrained to a single thread using `OMP_NUM_THREADS=1` and `TOKENIZERS_PARALLELISM=false` in `settings.py`.

2. Persistent Disk Volume Setup

ChromaDB data and uploaded files reside under `data/`. On Render, a 1 GB persistent disk must be mounted at `data/` to ensure indexed documents and FastEmbed model cache survive application redeployments.

3. Offline Fallback and Hashing Embeddings

If fastembed fails to download model BAAI/bge-small-en-v1.5 due to network timeouts, set `EMBEDDING_BACKEND=hashing` in environment variables. The engine automatically switches to local HashingEmbeddings (1024 dimensions) without failing.

4. Queue Saturation Error ERR-503

When multiple document uploads occur simultaneously, the bounded worker queue in `ingestion_queue.py` may become saturated. In this scenario, the API responds with HTTP status 503 SERVICE UNAVAILABLE and detail message Ingestion queue is full.

5. File Upload Size Limits ERR-413

Files exceeding `MAX_UPLOAD_MB` (default 50 MB) trigger HTTP status 413 REQUEST ENTITY TOO LARGE. The uploaded temporary file is unlinked immediately to protect disk storage.